ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jan 2021 P1HR Worked Solutions

Jan 2021 | P1HR

25 questions ▪ question image followed by worked solution

PREPARED BY

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION

Q#: 1 Marks: 3

TOPIC: **Fractions**

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1 Show that $3\frac{1}{5} \times 1\frac{5}{6} = 5\frac{13}{15}$

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 3

TOPIC: **Fractions**

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PAST PAPER QUESTION

Q#: 2 Marks: 4

TOPIC: **Statistics Toolkit**

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2 Given that $a < b < c$

the four whole numbers a, a, b and c have

a mode of 7
a median of 8.5
a mean of 9

Work out the value of a, the value of b and the value of c.

a = ...

b = ...

c = ...

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 4

TOPIC: **Statistics Toolkit**

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No worked solution is saved yet.

PAST PAPER QUESTION Q#: 3 Marks: 4

TOPIC: **Solving Inequalities**

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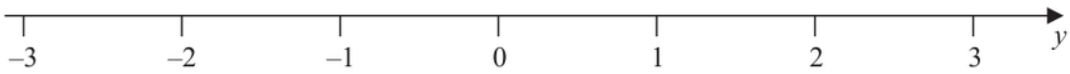
3 (a) On the number line, show the inequality $-2 \leqslant y < 1$



(2)

n is an integer.

(b) Write down all the values of n that satisfy $-3.4 < n \leqslant 2$

....

(2)

(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 4

TOPIC: **Solving Inequalities**

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 4 Marks: 3

TOPIC: **Standard & Compound Units**

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4 A train journey from Paris to Amsterdam took 3 hours 24 minutes.
The total distance the train travelled was 433.5 km.

Work out the average speed of the train.
Give your answer in kilometres per hour.

.... km/h

(Total for Question 4 is 3 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 3

TOPIC: **Standard & Compound Units**

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No worked solution is saved yet.

PAST PAPER QUESTION Q#: 5 Marks: 4

TOPIC: **Volume & Surface Area**

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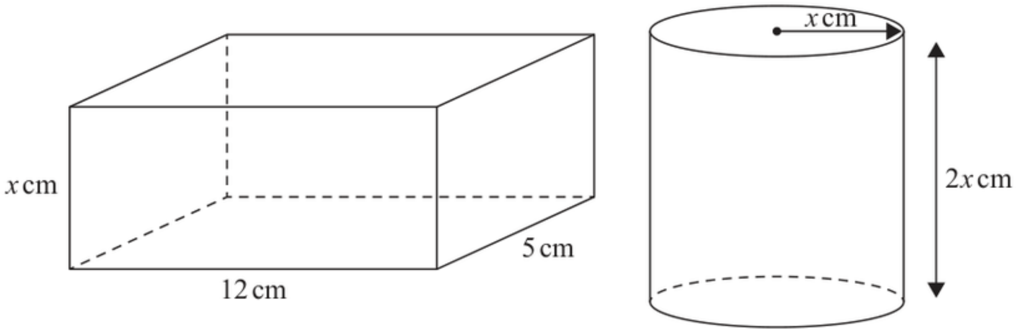
5 The diagram shows a cuboid and a cylinder.

Diagram **NOT** accurately drawn



The dimensions of the cuboid are x cm by 12 cm by 5 cm.
The volume of the cuboid is 270 cm^3

The radius of the cylinder is x cm.
The height of the cylinder is $2x$ cm.

(a) Work out the volume of the cylinder.
Give your answer correct to the nearest whole number.

.... cm^3
(3)

(b) Change 1 m^3 to cm^3

.... cm^3
(1)

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 4

TOPIC: **Volume & Surface Area**

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No worked solution is saved yet.

PAST PAPER QUESTION Q#: 6 Marks: 5

TOPIC: **Rearranging Formulas**

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6 (a) Make c the subject of $A = \dfrac{c}{y} - 5z$

....

(2)

(b) Write down the value of g^0

....

(1)

(c) Factorise $x^2 - 11x + 24$

....

(2)

(Total for Question 6 is 5 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 5

TOPIC: **Rearranging Formulas**

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No worked solution is saved yet.

PAST PAPER QUESTION Q#: 7 Marks: 3

TOPIC: **Compound Interest & Depreciation**

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7 Kuro invests 50 000 yen for 3 years in a savings account.
She gets 2.4% per year compound interest.

Work out how much money Kuro will have in her savings account at the end of the 3 years.
Give your answer correct to the nearest yen.

.... yen

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 3

TOPIC: **Compound Interest & Depreciation**

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 8 Marks: 5

TOPIC: **Angles in Polygons & Parallel Lines**

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8 The diagram shows a regular pentagon, *ABCDE*, a regular hexagon, *CFGHID*, and a quadrilateral, *EDIJ*.

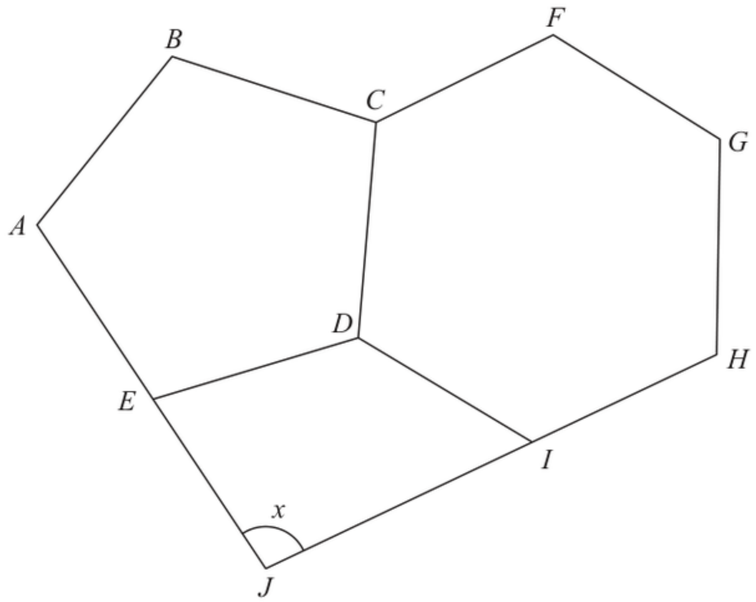

Diagram **NOT** accurately drawn

AEJ and *HIJ* are straight lines.

Work out the size of the angle marked *x*.
Show your working clearly.

.... °

(Total for Question 8 is 5 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 5

TOPIC: **Angles in Polygons & Parallel Lines**

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 9 Marks: 4

TOPIC: **Prime Factors, HCF & LCM**

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9 $A = 2^8 \times 3^5 \times 11^4 \qquad B = 2^6 \times 3 \times 11^8$

(a) Find the highest common factor (HCF) of A and B.

....

(2)

(b) Find the lowest common multiple (LCM) of $2A$ and $3B$.
Give the LCM as a product of powers of its prime factors.

....

(2)

(Total for Question 9 is 4 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 4

TOPIC: **Prime Factors, HCF & LCM**

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 10 Marks: 5

TOPIC: **Area & Perimeter**

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10 The diagram shows one face of a wall.
This face is in the shape of a pentagon with exactly one line of symmetry.

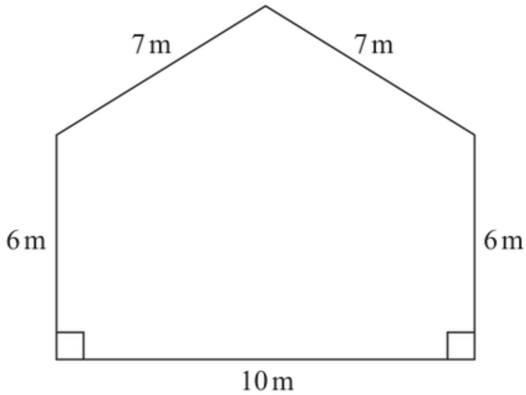

Diagram **NOT** accurately drawn

Omondi is going to paint this face of the wall once.
He has to buy all the paint that he needs to use.

The paint in each tin of paint Omondi is going to buy will cover $16\,\text{m}^2$ of the face of the wall.

Work out the least number of tins of paint Omondi will need to buy.
Show your working clearly.

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 5

TOPIC: **Area & Perimeter**

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 11 Marks: 6

TOPIC: **Cumulative Frequency Diagrams**

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11 The manager of a call centre asked the 120 people, who rang the call centre last week, how long they each waited before their call was answered.

The table gives information about their replies.

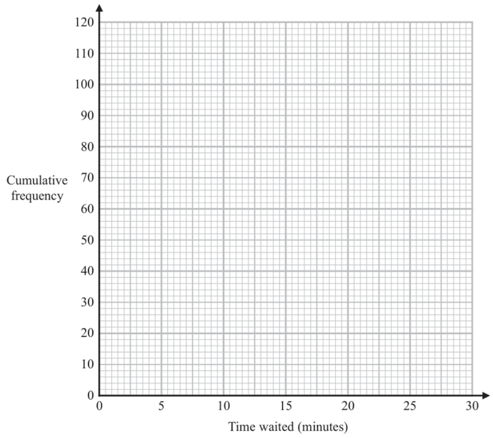
Time waited (t minutes)	Frequency
$0 < t \leqslant 5$	8
$5 < t \leqslant 10$	15
$10 < t \leqslant 15$	17
$15 < t \leqslant 20$	28
$20 < t \leqslant 25$	33
$25 < t \leqslant 30$	19

(a) Complete the cumulative frequency table.

Time waited (t minutes)	Cumulative frequency
$0 < t \leqslant 5$	
$0 < t \leqslant 10$	
$0 < t \leqslant 15$	
$0 < t \leqslant 20$	
$0 < t \leqslant 25$	
$0 < t \leqslant 30$	

(1)

(b) On the grid below, draw a cumulative frequency graph for your table.



(2)

(c) Use your graph to find an estimate for the median of the times waited.

.... minutes

(1)

(d) Using your graph, find an estimate for the percentage of the 120 people who said that they waited longer than 23 minutes before their call was answered.
Show your working clearly.

.... %

(2)

(Total for Question 11 is 6 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 6

TOPIC: **Cumulative Frequency Diagrams**

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PAST PAPER QUESTION

Q#: 12 Marks: 9

TOPIC: **Algebraic Roots & Indices**

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12 (a) Simplify $(16e^{10}f^{6})^{\frac{1}{2}}$

....
(2)

(b) Write $\dfrac{2x+1}{4}+\dfrac{x-2}{3}$ as a single fraction in its simplest form.

....
(3)

Given that $4^{k+3} = 16 \times 2^{k}$

(c) find the value of k.
Show your working clearly.

$k =$ ....
(4)

(Total for Question 12 is 9 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 9

TOPIC: **Algebraic Roots & Indices**

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PAST PAPER QUESTION

Q#: 13 Marks: 2

TOPIC: **Vectors**

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13 Here are two vectors.

$$\overrightarrow{AB} = \begin{pmatrix} 5 \\ 3 \end{pmatrix} \qquad \overrightarrow{CB} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$$

Find, as a column vector, $\overrightarrow{AC}$

(Total for Question 13 is 2 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 2

TOPIC: **Vectors**

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PAST PAPER QUESTION

Q#: 14 Marks: 4

TOPIC: **Circle Theorems**

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14

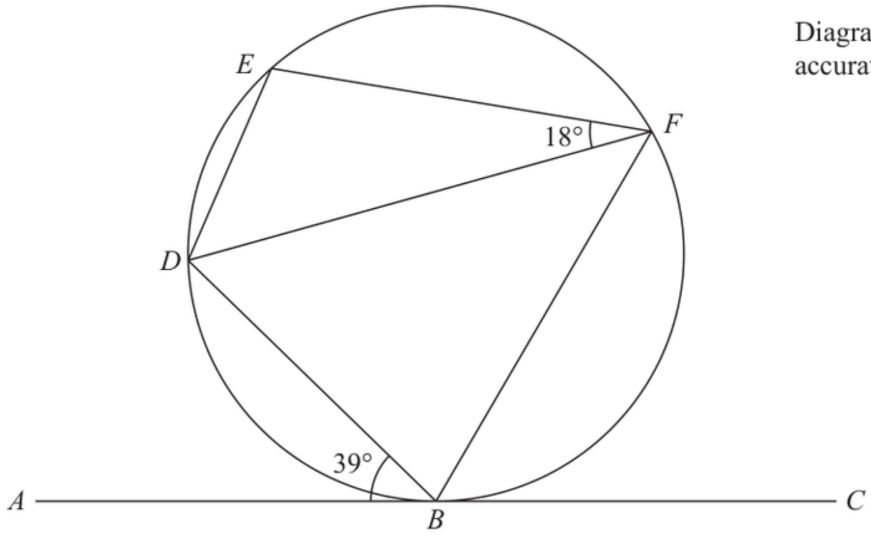

Diagram **NOT** accurately drawn

B, D, E and F are points on a circle.

ABC is the tangent at B to the circle.
Angle $ABD = 39°$
Angle $EFD = 18°$

Work out the size of angle BDE.
Give reasons for your working.

.... °

(Total for Question 14 is 4 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 4

TOPIC: **Circle Theorems**

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PAST PAPER QUESTION

TOPIC: **Algebraic Proof**

Jan 2021 P1HR - Jan 2021 | P1HR

Q#: 15 Marks: 5

15 (a) Use algebra to show that $4.\dot{5}\dot{7} = 4\frac{19}{33}$

(2)

(b) Show that $\dfrac{2}{6-3\sqrt{2}}$ can be written in the form $\dfrac{a+\sqrt{a}}{b}$

where a and b are integers.
Show your working clearly.

(3)

(Total for Question 15 is 5 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 5

TOPIC: **Algebraic Proof**

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PAST PAPER QUESTION

Q#: 16 Marks: 5

TOPIC: **Expanding Brackets**

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16 (a) Expand and simplify $(x + 4)(x - 2)(x + 1)$

....

(3)

(b) Express $x^2 - 10x + 40$ in the form $(x + a)^2 + b$, where a and b are integers.

....

(2)

(Total for Question 16 is 5 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 5

TOPIC: **Expanding Brackets**

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PAST PAPER QUESTION

Q#: 17 Marks: 5

TOPIC: **Simultaneous Equations**

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17 Solve the simultaneous equations

$$x - 6y = 5$$
$$xy - 2y^2 = 6$$

Show clear algebraic working.

(Total for Question 17 is 5 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 5

TOPIC: **Simultaneous Equations**

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WORKED SOLUTION

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PAST PAPER QUESTION

Q#: 18 Marks: 5

TOPIC: **Histograms**

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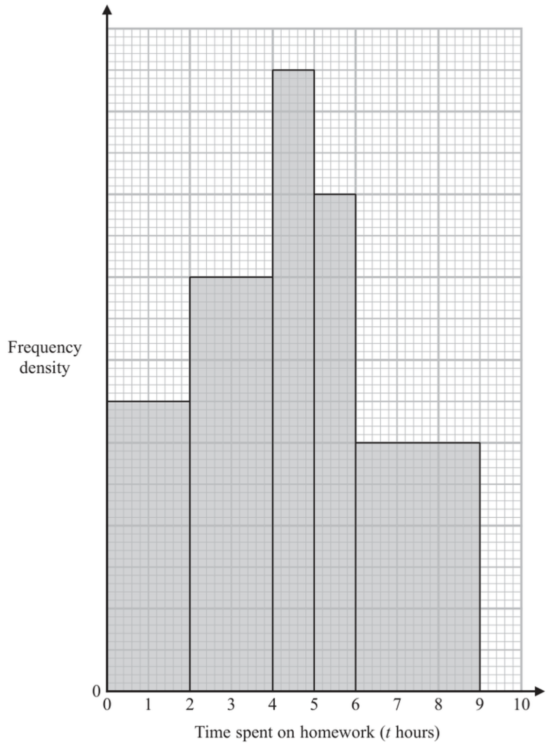
18 The histogram and the table give some information about the amounts of time, in hours, that Year 11 students at Bergdesh Academy spent, in total, on their homework last week.
No student in Year 11 spent longer than 9 hours on their homework.



Time spent on homework (t hours)	Frequency
$0 < t \leqslant 2$	28
$2 < t \leqslant 4$	
$4 < t \leqslant 5$	
$5 < t \leqslant 6$	
$6 < t \leqslant 9$	

Using the information in the histogram and in the table, work out an estimate for the mean amount of time the Year 11 students spent on their homework last week.
Give your answer in hours correct to 3 significant figures.

.... hours

(Total for Question 18 is 5 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 5

TOPIC: **Histograms**

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 19 Marks: 3

Topic: **Rounding, Estimation & Bounds**

Jan 2021 P1HR - Jan 2021 | P1HR

19 $k = \dfrac{t}{a - h}$

$t = 14$ correct to 2 significant figures
$a = 7.8$ correct to 2 significant figures
$h = 3.4$ correct to 2 significant figures

Work out the lower bound for the value of k.
Show your working clearly.

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 20 Marks: 5

TOPIC: **Differentiation**

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20 A particle P is moving along a straight line.
The fixed point O lies on the line.

At time t seconds ($t \geqslant 0$), the displacement of P from O is s metres where

$$s = t^3 - 9t^2 + 33t - 6$$

Find the minimum speed of P.

.... m/s

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 5

TOPIC: **Differentiation**

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WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 20 Marks: 5

TOPIC: **Differentiation**

Jan 2021 P1HR - Jan 2021 | P1HR

20 A particle P is moving along a straight line.
The fixed point O lies on the line.

At time t seconds ($t \geqslant 0$), the displacement of P from O is s metres where

$$s = t^3 - 9t^2 + 33t - 6$$

Find the minimum speed of P.

.... m/s

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 5

TOPIC: **Differentiation**

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PAST PAPER QUESTION

Q#: 21 Marks: 6

Topic: **Sequences**

Jan 2021 P1HR - Jan 2021 | P1HR

21 The nth term of an arithmetic series is u_n where $u_n > 0$ for all n
The sum to n terms of the series is S_n

Given that $u_4 = 6$ and that $S_{11} = (u_6)^2 + 18$

find the value of u_{20}

(Total for Question 21 is 6 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 6

TOPIC: **Sequences**

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WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION Q#: 21 Marks: 6

TOPIC: **Sequences**

Jan 2021 P1HR - Jan 2021 | P1HR

21 The nth term of an arithmetic series is u_n where $u_n > 0$ for all n
The sum to n terms of the series is S_n

Given that $u_4 = 6$ and that $S_{11} = (u_6)^2 + 18$

find the value of u_{20}

(Total for Question 21 is 6 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 6

TOPIC: **Sequences**

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No worked solution is saved yet.

PAST PAPER QUESTION Q#: 22 Marks: 5

TOPIC: **Coordinate Geometry**

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22 *ABC* is an isosceles triangle with $AB = AC$.

B is the point with coordinates (–1, 5)
C is the point with coordinates (2, 10)
M is the midpoint of *BC*.

Find an equation of the line through the points *A* and *M*.
Give your answer in the form $py + qx = r$ where p, q and r are integers.

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Coordinate Geometry**

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PAST PAPER QUESTION

Topic: **Coordinate Geometry**

Jan 2021 P1HR - Jan 2021 | P1HR

Q#: 22 Marks: 5

22 *ABC* is an isosceles triangle with $AB = AC$.

B is the point with coordinates (–1, 5)
C is the point with coordinates (2, 10)
M is the midpoint of *BC*.

Find an equation of the line through the points *A* and *M*.
Give your answer in the form $py + qx = r$ where p, q and r are integers.

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Coordinate Geometry**

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